FYP – OpenPaths

Matthew Pirotta

Supervisor: Dr. Kristian Guillaumier

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L-Università ta' Malta
Faculty of Information & Communication Technology

Contents

List of Acronyms

ABM Agent-Based Modelling.

AT-ABM Active Transportation Agent-Based Model.

BNDP Bicycle Network Design Problem.

C-SAM Connections for Safer Active Mobility.

CC Closest Component.

DNDP Discrete Network Design Problem.

L2C Largest-to-Closest.

L2S Largest-to-Second.

LP Linear Programming.

MILP Mixed Integer Linear Programming.

MPAM Multi-Class Pedestrian Assignment Model.

NHTS National Household Travel Survey.

NSGA-II Non-Dominated Sorting Genetic Algorithm II.

NSO National Statistics Office.

OD Origin–Destination.

OSM Open Street Map.

POI point of interest.

R2C Random to Closest.

SPSA Simultaneous Perturbation Stochastic Approximation.

TAZ Traffic Assignment Zones.

TOPSIS Technique for Order Preference by Similarity to Ideal Solution.

UNDP Urban Network Design Problem.

1 Introduction

Transportation systems influence multiple sectors of society, shaping the environment, public health, and socioeconomic relationships. Decades of car-centric urban development have contributed to rising carbon emissions, escalating road maintenance costs, and the creation of hostile public spaces that discourage active mobility.

A substantial portion of national expenditure is directed toward road construction and maintenance. According to CEIC data [1], Malta spent a median of approximately €21 million annually between 1995 and 2014. Despite these investments, road safety remains a significant issue. National Statistics Office (NSO) [2] reported around 16,000 road accidents in 2023, including 26 incidents involving cyclists, eight of whom sustained grievous injuries. Such figures highlight both the financial and societal costs of car dependency for the country as well as the inherent dangers to cyclists. All of this is further reinforced by the 2030 National Transport Master Plan, which estimates that traffic congestion will cost Malta €770 million in 2025, rising to €917 million annually by 2030 [3]. The report attributes these losses to longer journey times, higher vehicle operating costs, and Malta's heavy reliance on private cars, which account for over 84% of all road traffic. It further warns that congestion already costs the country around 3.3% of its GDP, excluding environmental externalities such as emissions and pollution.

Importantly, the same report identifies active mobility, such as cycling and walking, as part of the solution to ease congestion, yet acknowledges that Malta's cycling network remains "fragmented" and "does not penetrate urban areas," with infrastructure that is often "poorly designed" despite recent improvements [3]. This highlights a significant gap between the policy vision for sustainable transport and the current state of infrastructure on the ground.

Maltese citizens do not see the country as being apt to cycle in, as reflected in the National Household Travel Survey (NHTS) of 2021 [4], which reported that only 0.7% of all trips were carried out by bicycle. Respondents identified road safety concerns (17%) and poor-quality cycling infrastructure (13%) as the principal barriers to cycling. The Maltese government has shown commitment to improve the local cycling infrastructure in the recent years, with the ongoing project "*Vjal Kulhadd*" launched in 2024 [5], with an investment of 10 million euros [6]. A segregated passageway dedicated to pedestrians and cyclists from Żabbar to Żejtun was completed in March 2025 [7], with the first phase of this project planned to be completed by 2026. Another project is the Connections for Safer Active Mobility (C-SAM) initiative, which is a nationwide cycling and active-mobility network plan for Malta announced in 2022, with an estimated budget of €35 million and a target of 50-60 km of new cycling routes [8]. The Grand Harbour Cycling Network is the first component of C-SAM with a budget of €27.4 million, focusing on the Grand Harbour area (encompassing Msida, Pietà, Blata l-Bajda, Valletta, Marsa and Paola) and is scheduled for completion in phases by 2028 [9].

In parallel, government initiatives are also targeting public transport improvements to reduce reliance on private vehicles. These include the addition of 25 new buses in 2025 [10] and renewed discussions around the proposed Malta Metro system, initially announced in 2021 [11], temporarily shelved, and later revived for consultation in 2025 [12].

Together, these efforts reflect a growing recognition of the need for a balanced and sustainable mobility system, where active and public transport play central roles in shaping Malta's urban future. Transitioning cities toward walkable and cyclable environments can improve public health, reduce emissions, and foster safer and more inclusive urban spaces. To support this transition to a more expansive cycling network, there is a growing need for data-driven tools that assist urban planners in reallocating road space from private cars to active mobility users.

The overarching aim of this project is therefore to develop a computational framework that optimally expands Malta's cycling network, ensuring safe and continuous connections between major destinations while maintaining a functional and connected road system for cars. The current transport context in Malta underscores the urgency of this objective.

2 Literature Review

Designing efficient and safe bicycle networks requires an understanding of how transport systems can be represented, analysed, and optimised using quantitative models. The literature on network design spans a range of approaches that variously emphasise network structure, infrastructure optimisation, and traveller behaviour. Reflecting this diversity, this review first outlines the fundamental concepts common to all modelling frameworks, including graph representation, flow estimation, and network evaluation. Then examining the main optimisation approaches used in bicycle network design: heuristic, linear programming and mixed-integer linear programming, and metaheuristic

2.1 Fundamental Concepts in Road Network Modelling

2.1.1 Graph Representation of Transport Networks

All network design frameworks represent the underlying road infrastructure as a graph, where intersections correspond to nodes and road segments to edges. Across the reviewed studies, network representation varies according to modelling objective. Heuristic or topological studies e.g [13] adopt undirected simple graphs to focus on geometric growth and network efficiency rather than directional flow. In contrast, most optimization-based approaches employ directed graphs, which are the most common representation in transport network modelling because they capture one-way streets and allow for direction level distinction of features, providing a realistic basis for traffic assignment. More advanced frameworks, such as [14], extend this to directed multigraphs that allow lane-level differentiation, where each lane is represented as its own directed edge. Offering finer control over lane-specific attributes such as capacity and flow allocation. This distinction highlights the trade-off between computational simplicity and flow realism in urban network design.

An important extension of this graph representation is the use of multilayer networks, which capture the interactions between different modes of transport. In such models, each layer corresponds to a distinct mode, such as motorised traffic, cycling, or walking, while shared intersections or transfer points act as inter-layer connections. This representation allows planners to account for modal dependencies and conflicts, for example when cycling lanes share space with car traffic or when cyclists use protected corridors separated from the vehicular network. Lonardi et al. [15] explicitly adopt this formulation in their framework, modelling the urban transport system as coupled road and protected-cycle layers to evaluate how improvements in one layer affect the other. Even studies that focus solely on the cycling network implicitly rely on this concept, as proposed bicycle links are embedded within an existing multilayered urban fabric that includes vehicle and pedestrian flows.

Such approaches are examples of the broader Urban Network Design Problem (UNDP), and more specifically the Discrete Network Design Problem (DNDP), which is generally considered NP-hard [14]. This complexity arises because network design involves bi-level decision-making. At the upper level, discrete infrastructure choices are made, such as reallocating or upgrading road segments. At the lower level, user flows then adjust to the new network conditions. The interaction between these two levels produces a non-convex and combinatorial search space, making it computationally intractable to find the global optimum for large-scale networks. The way travel demand and flow are represented at this lower level directly affects both the realism of user responses and the computational tractability of the overall optimisation. Accurately and efficiently capturing this interaction is therefore essential for evaluating how proposed network changes influence travel behaviour and system performance.

2.1.2 Flow and Demand Representation

Representing travel demand and flow accurately is a critical element of transport network modelling. In essence, while the graph defines the spatial structure of the transport system, flow representation determines how travellers use that structure, that is, how many trips occur between different locations and how these trips distribute across available routes. There are two main models capturing this relationship at varying levels of detail, Origin–Destination (OD) matrices and trajectory data.

An OD matrix defines the total number of trips between each origin and destination pair within the study area, usually derived from household travel surveys or census statistics. OD data provides a macroscopic view of mobility patterns, making it suitable for equilibrium-based and aggregate network models where flow assignment depends on total demand rather than individual behaviour. For instance, [16, 17] both rely on OD data to simulate pedestrian and cycling flows at a city scale. Another instance, in the Tehran case study, [16] estimated the OD matrix using a combination of household surveys and field observations, such as counting pedestrians at subway stations and bus stops. Trips were classified into home-based and non-home-based categories, further subdivided into work and leisure purposes, allowing the model to distinguish between trip types with different sensitivities to distance and environmental factors. In contrast [13] assume uniform OD demand between all node pairs to isolate the topological effects of network structure.

On the other hand, trajectory data such as GPS traces, or crowdsourced cycling data (e.g., Strava), captures individual-level travel behaviour with high spatial and temporal precision. This data enables the reconstruction of actual routes taken by users, revealing preferences for safety, directness, or comfort that OD matrices cannot capture. For example, [18] employ bike trajectory data to identify observed cyclist movement patterns and use these trajectories to guide the placement of new cycling links, thereby improving the behavioural realism of their model. Trajectory datasets are therefore particularly valuable for agent-based models, where individual route choices and responses to infrastructure are explicitly modelled.

Each data source presents trade-offs. Trajectory data provides detailed behavioural information but often suffers from sampling bias, such as overrepresentation of fitness-oriented or app-using cyclists and privacy limitations, while OD matrices are more systematically collected but lack route-level detail and temporal variation.

Once travel demand and flows are defined through OD matrices or trajectory data, the next step in evaluating bicycle networks is to quantify their structural and functional performance using a range of network metrics or agent based modelling.

2.1.3 Network Evaluation Frameworks

Topological Metrics Network metrics provide quantitative indicators of how effectively a transport network supports mobility, accessibility, and resilience. These metrics are derived from graph theory and in literature are used both to evaluate existing networks and to guide heuristic or optimization models toward more efficient configurations.

Connectedness is among the most fundamental measures, describing whether the network forms a single, navigable system or remains fragmented into multiple components. Metrics such as the size of the largest connected component or the number of disconnected subgraphs are commonly used to assess overall network cohesion. High connectedness indicates that users can reach most destinations without detouring the safe cycle network, an essential property for promoting cycling continuity and accessibility.

Directness measures how efficiently users can travel between two points and is typically defined as the ratio of Euclidean distance to shortest-path distance within the network. Natera et al. [19] take a different approach to directness, calculating it as the ratio of distance travelling with the car compared to the distance travelling with the bicycle. Values closer to one indicate straighter, more direct routes and fewer unnecessary detours, important for both user comfort and competitiveness with motorised travel. Studies such as [13] use directness to evaluate how heuristic growth strategies improve route efficiency as new segments are added.

Centrality metrics capture the relative importance of nodes or links within the network. Betweenness centrality, for example, measures how frequently a link lies on the shortest paths between all node pairs, serving as a proxy for potential flow or load. High-betweenness edges often correspond to corridors that would carry the most traffic if demand were evenly distributed, making this measure a key input in centrality-based heuristics [13]. Other centrality measures, such as closeness (proximity of a node to all others), and degree (number of connections per node), indicate network accessibility and local connectivity density.

Beyond these classical indicators, Szell et al. [13] employ higher-order measures such as global and local efficiency, spatial clustering, anisotropy, and coverage. These capture the balance between resilience, compactness, directional bias, and service reach. For instance, global efficiency measures the overall accessibility of a network by quantifying how efficiently any two nodes can reach each other through the existing connections, while local efficiency evaluates how well a neighbourhood remains connected if a node fails. Spatial clustering and anisotropy help identify concentration patterns and directional preferences, offering insights into the morphological coherence of urban cycling networks. Lastly coverage offers a reasonable buffer where commuters can reach getting out/in the bike network. Unlike purely geometric measures, perceived safety captures how infrastructure design, such as physical separation, traffic volume, or intersection treatment, affects the attractiveness and usability of cycling routes, providing a human-centred dimension to network evaluation.

Together, these metrics provide a quantitative framework for assessing and comparing networks across cities or growth strategies. They are particularly useful for heuristic and optimization approaches that aim to improve network structure without modelling detailed individual behaviour. However, as the next section discusses, agent-based models complement these static indicators by capturing the dynamic, individual-level interactions that give rise to aggregate network flows.

Agent-Based and Activity-Based Behavioural Modelling Agent-Based Modelling (ABM) and activity-based behavioural models are primarily concerned with how travellers make choices under changing transport conditions. Their central purpose is to represent decision-making processes such as mode choice, route choice, departure-time choice, and behavioural adaptation, allowing aggregate travel patterns to emerge from the actions of heterogeneous individuals. In the context of active transportation, this makes them especially useful for examining how changes in cycling and walking infrastructure alter safety perceptions, travel preferences, and the likelihood of modal shift. Unlike more aggregate or topological approaches, these models explicitly encode person-level decision rules and interactions with the surrounding urban environment. Their value therefore lies less in reproducing the detailed operational performance of a specific network and more in explaining how infrastructure change influences behaviour. At the same time, the credibility of their outputs depends heavily on data quality, calibration, and the realism of the behavioural assumptions embedded in the model.

Agent-based evaluation methods vary widely in their spatial scale, behavioural detail, and underlying data sources. Aziz et al. [20] developed the Active Transportation Agent-Based Model (AT-ABM) to simulate neighbourhood-scale mode choice between home and work trips. In this model, the agents' baseline probabilities of selecting a transport mode depend on attributes such as income, age, and perceived safety, and are further influenced by social interactions with other agents within the same local environment. The model integrates traffic safety records, capturing pedestrian and cyclist crash frequencies with population and socio-demographic datasets to parameterise agent characteristics. Because the AT-ABM is non-deterministic and lacks analytical gradients, it employs the Simultaneous Perturbation Stochastic Approximation (SPSA) algorithm for heuristic calibration, iteratively perturbing model parameters and evaluating outcome changes to efficiently estimate behavioural sensitivities under stochastic variation.

In contrast, [21] adopt a rule-based ABM to explore how the introduction of separated cycling infrastructure affects driver behaviour and road safety. Using a network of approximately 13,000 road segments, the model represents behavioural adaptation in which drivers gradually learn to anticipate cyclists after repeated encounters, leading to reduced collision risks up to a saturation point. Road types are differentiated by exposure and visibility levels, and potential crash events are simulated at intersections based on whether drivers have recently encountered cyclists. Rather than relying on statistical calibration, parameters such as driver learning rate and perception sensitivity are systematically varied across scenarios to examine emergent safety dynamics. This approach provides a dynamic behavioural perspective absent from traditional static safety models, illustrating how ABMs can capture the temporal evolution of driver awareness and risk perception.

A broader, city-scale application is demonstrated by Jafari et al. [22], who employ an integrated agent-and-activity-based simulation framework to assess changes in travel mode share resulting from proposed cycling infrastructure in Melbourne, Australia. The model makes use of census-based demographic data, OD matrices, and household travel surveys to define behavioural clusters of riders characterised by trip length, safety preference, and socio-economic background. These clusters inform the agent population, allowing simulation of how a proposed cycle corridor would alter travel behaviour across demographic groups. Calibration is achieved deterministically by iteratively adjusting mode-choice constants until simulated mode shares align with observed data. The results show that travel time strongly influences mode choice across all clusters, and that improved cycling infrastructure and perceived safety substantially increase cycling uptake, by approximately 30%, highlighting the value of agent-based approaches for infrastructure impact assessment and policy evaluation.

Collectively, these studies show the value of agent-based and activity-based approaches for modelling travel choice in active transportation systems. By representing the decision-making processes of individual travellers, these models help explain how and why infrastructure changes influence user choices, safety outcomes, and modal shifts, moving beyond purely structural or geometric assessments of the network. Their flexibility makes them particularly useful for studying behavioural adaptation, social influence, and other mechanisms through which transport preferences evolve over time.

Nevertheless, several methodological challenges remain. These models are data-intensive, requiring detailed demographic, safety, and travel diary information to parameterise and validate behavioural rules. Their complexity introduces large numbers of parameters, making calibration and validation difficult. Models such as that of Aziz et al. [20] address this through stochastic optimisation techniques like SPSA, while others, such as Thompson et al. [21], rely on heuristic or manual parameter variation, potentially affecting reproducibility and generalisability. Moreover, many such models are stronger at representing mode choice and aggregate travel response than at reproducing detailed route-level operations on a specific network. As a result, they are well suited to analysing demand adaptation and behavioural change, but less directly suited to evaluating junction performance, queue spillback, or other operational effects of a particular street redesign.

2.2 Microscopic Network Simulation

Microscopic network simulation addresses a different question. Rather than modelling how travellers form preferences, it models how individual road users move through an explicitly specified transport network under given demand, control, and infrastructure conditions. Its main strength is therefore operational rather than behavioural: it is used to analyse congestion, delay, queueing, bottlenecks, and route-level interactions that emerge from the geometry and control of the network itself.

Microscopic traffic simulation is widely used in transport research to examine how changes in road layout, control, and allocation of street space affect traffic operations. Its main advantage is that it represents traffic at the level of individual road users moving through an explicit network, making it suitable for analysing congestion, delay, queue formation, and route-level interactions under alternative infrastructure scenarios. Several software platforms are commonly used for this purpose, including SUMO, MATSim, PTV Vissim, and Aimsun Next, although they differ in modelling emphasis and typical application [23]. SUMO is an open-source, microscopic, continuous, and multimodal simulator designed for large networks. MATSim is an agent-based transport simulation framework in which travellers iteratively adapt their plans. PTV Vissim is oriented toward highly detailed microscopic operational modelling, while Aimsun Next supports microscopic, mesoscopic, and hybrid simulation together with dynamic traffic assignment.

In broad terms, these tools occupy different positions within the transport-modelling landscape. MATSim is typically associated with activity-based and agent-based analysis of travel behaviour and demand adaptation, whereas Vissim and Aimsun are often used for detailed operational studies of network performance and control. SUMO has become especially prominent in academic work because it combines microscopic network simulation with an open and extensible workflow [24]. This combination has made it widely used in research on traffic operations, multimodal interaction, and scenario testing. The recent systematic review *Agent-Based Modeling in Urban Human Mobility: A Systematic Review* [25] identifies SUMO as the most frequently used ABM tool in the reviewed literature, accounting for 26.13% of identified applications, ahead of AnyLogic, NetLogo, GAMA, and MATSim.

Comparative studies also show that these platforms involve different trade-offs between realism, scale, and flexibility. In *A Comparison of Microscopic Traffic Flow Simulation Systems for an Urban Area* [24], Maciejewski compares TRANSIMS, SUMO, and VISSIM on a real urban network and characterises SUMO as a space-continuous microscopic simulator that offers a practical compromise between modelling detail and scalability. The paper contrasts this with VISSIM's stronger geometric precision for complex intersections and TRANSIMS' coarser but highly scalable cellular-automata approach. The same comparison also notes that SUMO supports route generation from turning ratios and is applicable to city- and region-scale networks, which helps explain its continued use in urban network studies.

This literature is particularly relevant for studies of cycling-infrastructure reallocation. Recent work such as *Impact of Radical Bike Lane Allocation on Bi-Modal Urban Road Network Traffic Performance: A Simulation Case Study* [26] uses SUMO to evaluate how reallocating road space to dedicated bicycle infrastructure affects both car traffic and bicycle traffic. That study analyses origin–destination travel times, congestion development, and network-level traffic states under alternative car–bike modal shares, illustrating how microscopic simulation can be used to assess the operational consequences of reallocating street space between modes.

Taken together, the literature suggests that simulation software should be understood less as a hierarchy of better or worse tools than as a set of platforms with different analytical strengths. Agent-based frameworks are especially useful when the focus is long-run behavioural adaptation and travel-demand response, whereas microscopic traffic simulators are particularly well suited to evaluating the short-run operational effects of changes in network design. Within that latter category, SUMO appears frequently in the literature and is widely used because it provides microscopic, multimodal, and network-explicit simulation in an open research environment.

These evaluation frameworks provide the means to assess and simulate how effectively a network supports mobility and behavioural adaptation. However, evaluation alone does not determine which infrastructure changes should be implemented. The next challenge is therefore optimisation, identifying from the many possible interventions, the set of new segments or reallocations that most improve overall network performance. The following section reviews the main optimisation approaches proposed for this purpose.

2.3 Optimisation Approaches for Network Design

In the literature, the upper-level optimization problem in bicycle network design is typically addressed through one of three main approaches: heuristic, mathematical programming, and metaheuristic methods. Each approach presents distinct advantages, limitations, and suitable use cases, balancing between computational efficiency, behavioural realism, and scalability. With this foundation on how these problems are conceptually structured, the following sections discuss the different modelling approaches in greater detail.

2.3.1 Heuristic Approaches

The simplest of the methods used in bicycle network planning are heuristic approaches. These are considered "simple" because they rely on rule-based or greedy decision-making guided by intuitive network metrics such as connectedness, directness, or betweenness centrality. Heuristic models iteratively add or remove edges in the bicycle network to improve a selected topological metric, without formulating the problem as a formal bi-level optimization task. In doing so, they make several simplifying assumptions, most notably, that traffic demand is uniformly distributed across the network and that travellers do not adjust their routes in response to infrastructure changes (i.e., no user equilibrium is modelled). As a result, these methods capture the geometric and structural characteristics of a network but neglect behavioural feedback and flow dynamics.

For instance, Szell et al. [13] adopt a centrality-based heuristic, where new links are added according to graph-theoretic importance measures, such as betweenness or closeness centrality. This approach prioritizes edges that are topologically critical or spatially central within the underlying street network, allowing the simulated bicycle network to grow along corridors that would most efficiently shorten travel distances.

In contrast, Natera et al. [19] propose component-based heuristics designed to reconnect fragmented pieces of existing infrastructure. Their Largest-to-Second (L2S)strategy links the two largest disconnected components to quickly consolidate the network, while the Largest-to-Closest (L2C) strategy connects the largest component to its nearest neighbour, favouring shorter and more spatially efficient connections. Both approaches share the same principle of incremental, topology-driven expansion, but differ in how they define the "importance" of the next edge to add: L2S biases towards overall connectedness, whereas L2C favours directness by preferring shorter spatial gaps. The authors also use simpler baselines for comparison, using Random to Closest (R2C), representing uncoordinated development, and Closest Component (CC), which connects the nearest two subgraphs, producing a minium spanning tree that minimises investment but is not resilient to any interruptions. Overall, L2C performed best, with L2S also strong but slightly less effective, both outperforming the baselines. The study shows that targeted investments into key missing links can rapidly consolidate fragmented bicycle networks, though cars still achieve greater directness, with routes averaging about 33% shorter than those of bicycles.

A third paradigm is introduced by Steinacker et al. [27] applies percolation-based heuristics to model network growth and robustness over time. In a forward percolation approach, the network gradually expands by adding segments with the highest marginal contribution to global accessibility or welfare. Conversely, the inverse percolation approach begins from a fully built network and iteratively removes the least critical segments, recalculating connectivity and demand after each step. An additional advantage of the inverse percolation method is that it inherently maintains full network connectivity throughout the process, as segments are only removed if their deletion does not fragment the network. This ensures that all resulting configurations remain fully functional while progressively identifying the most critical corridors. These methods provide a transparent way to approximate how incremental investments or disinvestments affect overall network integrity.

Heuristic models consistently outperform baseline random growth strategies. Centrality-based growth yields higher connectedness, directness, and resilience than both random and minimum spanning tree baselines, producing networks that more closely resemble those of well-developed cycling cities such as Copenhagen [13]. Component-based heuristics (L2S and L2C) outperform random or closest-component baselines by achieving faster consolidation and greater overall cohesion [19]. These findings demonstrate that, even under uniform demand assumptions, simple topology-driven heuristics can reproduce high-quality, cohesive bicycle networks far more efficiently than unguided or piecemeal local planning.

While all heuristic families improve structural aspects of the network, they share fundamental simplifying assumptions. All assume uniform travel demand across the city and neglect behavioural factors such as route choice, mode preference, or induced demand. Centrality-based growth interprets betweenness as a proxy for flow under uniform conditions, component-based heuristics connect infrastructure purely by geometry, and percolation-based methods evaluate robustness statically without modelling user adaptation. Consequently, heuristic approaches provide a computationally efficient and transparent means of exploring large-scale bicycle networks and identifying missing or critical links, but their simplicity comes at the cost of behavioural realism, dynamic feedback, and global optimality.

2.3.2 Flow-Based and Optimal-Transport Models

Optimal Transport (OT)–based models provide a continuous, physics-inspired framework for evaluating cyclist movement and network performance. Rather than simulating individual agents or relying solely on topological indicators, these models treat cyclist flows as mass transported along a network under cost-minimising principles. Lonardi et al. [15] formulate cyclist routing as an OT-driven dynamical system in which edge 'capacities' adapt over time to match observed fluxes. This allows the network to converge to weighted shortest-path flows that reflect cyclists' safety preferences and infrastructure coverage, effectively capturing how demand and infrastructure co-evolve.

Their analysis demonstrates how variations in network completeness affect global metrics such as detour, overlap, and congestion inequality, revealing substantial spatial disparities across Copenhagen's districts. While these OT models provide a fluid, adaptive view of network performance, they typically treat capacity as a continuous variable. To address the discrete 'build-or-not' decisions and strict budgetary constraints faced by city planners, researchers often turn to the more computationally intensive framework of Mathematical-Optimisation.

2.3.3 Mathematical-Optimisation Approaches

Mathematical-optimisation approaches including Linear Programming (LP), Mixed Integer Linear Programming (MILP), and related variants represent the most analytically rigorous class of methods for solving the Bicycle Network Design Problem (BNDP). These models treat network design as a constrained optimisation problem in which infrastructure decisions are chosen to minimise or maximise explicit objective functions (e.g., travel time, continuity, or welfare) while respecting physical, spatial, and budgetary constraints. Unlike heuristic approaches, which rely on structural rules or centrality metrics, optimisation-based methods search a well-defined feasible space and provide transparent trade-offs between competing goals such as cycling accessibility, car-network performance, construction costs, and equity considerations.

LP formulations optimise continuous variables, such as allocating a street's lane capacity between cyclists and cars. MILP extends this framework by introducing binary or integer decision variables that capture discrete choices, including whether a protected bike lane is installed on a link or whether an upgrade is built along a cyclist trajectory. This increases modelling realism but also substantially raises computational complexity. Large-scale MILPs are NP-hard, often requiring relaxation, decomposition, or iterative rounding techniques to remain tractable [14]. Furthermore, most LP/MILP models rely on static route choice and fixed OD matrices, assumptions that simplify computation but limit behavioural realism compared to agent-based or dynamic assignment models.

Several studies emphasise the computational intractability of bicycle network design. Wiedemann et al. [14] note that even subtasks of their model—such as optimising lane direction via edge flipping to maintain connectivity—are already NP-hard, implying that the full lane-allocation problem is at least equally difficult. Liu et al.'s [18] trajectory-based MILP further inherits NP-hardness through its binary knapsack link-selection variables, and Rastbin et al. [16] explicitly classify their bi-level pedestrian–cycle optimisation framework as NP-hard. Collectively, these works confirm that urban bicycle network design lies firmly within the NP-hard class, motivating the use of relaxations, heuristics, and metaheuristic algorithms to obtain tractable solutions for real-world networks.

Linear Programming Wiedemann et al. [14] propose a multi-criteria linear programming model that allocates limited street capacity between cars and cyclists. The model minimises the combined travel time of both modes, weighted by a tunable parameter that expresses the policy preference between car and bicycle performance. To reduce computational cost, the authors apply a linear relaxation of the integer lane-allocation problem and iteratively fix the highest-capacity edges after each run, progressively reducing the search space. The final model is able to handle 830-edge networks on a single core up to 10,000 edge networks using parallel processing. The approach produced a 35 % improvement in cycling performance with less than a 20 % increase in car travel time, outperforming heuristic baselines.However, the model tends to allocate one lane of each two-lane road to cyclists, neglecting detailed congestion dynamics for cars. Finally, the result took about 16 hours to reach.

Mixed-Integer Linear Programming The trajectory-based BL-AC and BL-GU models [18] formulate bicycle network design as a true MILP that integrates observed cyclist trajectories into the decision process. Each candidate road segment is associated with a binary construction variable, allowing the model to explicitly represent whether a proper bike lane is installed. BL-AC (Adjacency Continuity) maximises the number of pairwise-adjacent constructed segments, producing geographically dispersed upgrades with broad coverage. In contrast, BL-GU (Generalised Utility) builds longer, more cohesive corridors by maximising a supermodular utility function tied to the size of continuous constructed segments. The models incorporate travel-time disutility, cyclist preferences, and—when extended with a route-choice layer—Multinomial Logit (MNL) probabilities that account for how users may alter their routes in response to infrastructure changes. Empirically, BL-AC yields higher coverage but shorter continuous paths, whereas BL-GU produces substantially larger contiguous corridors (up to 49 connected segments compared to 21 in BL-AC). Including route choice causes infrastructure to geographically spread out rather than forming a single dominant spine, underscoring the importance of behavioural feedback in MILP-based planning. While computationally more demanding than LP relaxations, these models demonstrate how discrete link-level decisions and behavioural realism can be embedded into optimisation-based network design.

Related Mathematical-Optimisation Approaches Direct Optimisation Steinacker et al. [27] extend optimisation-based bicycle network design by introducing a temporal decision-making framework that determines not only where, but also when, segments of Copenhagen's cycle-superhighway network should be built. Unlike a single global MILP that optimises all years simultaneously, their method solves a sequence of independent annual optimisation problems, each selecting the next set of links that maximises time-discounted welfare (Net Present Value) given the infrastructure already constructed. This "direct optimisation" approach therefore yields year-by-year optimal decisions, but does not guarantee global optimality across the entire planning horizon. Instead, it provides a computationally tractable way to incorporate phasing, discounting, and evolving network conditions into the planning process. Within the broader landscape of LP/MILP-style models, Steinacker et al. [27] demonstrate how temporal, economic, and multimodal considerations can be embedded into optimisation frameworks without constructing a fully time-expanded MILP, which would be computationally prohibitive for decades-long horizons.

While Steinacker et al. [27] report that their direct-optimisation strategy yields marginally higher welfare than the heuristic benchmark, the principal differences arise in the temporal sequencing of construction rather than the overall extent of the resulting network. Their uncertainty analysis further shows that demand variation has minimal influence on welfare outcomes, whereas cost uncertainty substantially affects optimisation-driven plans.

Synthesis While these models guarantee mathematical optimality and offer interpretable policy trade-offs, their precision comes at a computational cost. Solving large-scale LPs\MILPs with spatial or behavioural constraints remains NP-hard, often requiring simplifications such as relaxations (temporarily allowing integer variables to take continuous values), decomposition (solving smaller subproblems), or iterative rounding (progressively fixing variables) to maintain tractability [14]. These simplifications improve computational efficiency but sacrifice global optimality, resulting in solutions that approximate the best network configuration rather than guaranteeing it. Moreover, most LP\MILP based formulations generally rely on simplified behavioural assumptions, for example, fixed OD demand, where the total number of trips between origins and destinations remains constant, and static route choice, where users follow a single optimal path that does not adjust to changing network conditions. These assumptions make MILP frameworks computationally efficient but limit their ability to capture behavioural feedbacks such as induced demand, route adaptation, or mode shifts, which are better represented in agent-based models.

Together, these optimisation-based methods demonstrate how linear, mixed-integer, temporal, and dynamical extensions enhance the policy relevance of bicycle network planning. LP and MILP formulations offer globally coordinated and interpretable solutions but rely on simplified behavioural models; temporal direct-optimisation frameworks incorporate economic phasing and uncertainty; Despite their limitations, optimisation-based approaches deliver transparent, reproducible, and competitive alternatives to heuristic methods, and continue to form the backbone of analytically rigorous bicycle network design research.

To address the computational challenges, research has explored metaheuristic optimization that can handle multi-objective problems more flexibly.

2.3.4 Metaheuristic Approaches

Metaheuristic approaches are a less explored class of methods that aims to balance realism and computational feasibility, modelling the real world more accurately than simple heuristics while avoiding the exhaustive search required by exact MILP formulations. To achieve this, [16, 17] employ the non-dominated Sorting Non-Dominated Sorting Genetic Algorithm II (NSGA-II) to solve multi-objective optimization problems that simultaneously consider conflicting objectives for non-motorist and car users. NSGA-II is an evolutionary algorithm that iteratively evolves a population of candidate solutions, ranking them based on their Pareto dominance i.e. how well they perform relative to others across multiple objectives without being strictly better or worse in all of them. The outcome is a Pareto front, a set of optimal trade-off solutions where improving one objective (e.g., cyclist safety or travel time) would necessarily worsen another (e.g., car travel efficiency). Within this front, crowding distance is used to preserve diversity among non-dominated solutions, preventing premature convergence to a single region of the search space. After convergence, the final non-dominated set is further refined using the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) method to rank and select the most balanced designs. This approach enables planners to explore a spectrum of balanced network configurations rather than a single, rigid optimum.

In Rastbin et al. [16] when evaluating the model on the large-scale case study on the historical-Cultural district of Tehran, Iran (approximately 2,000 hectares, 55 nodes, and 98 edges), the proposed NSGA-II bi-level model demonstrated strong computational efficiency and scalability. The lower-level Multi-Class Pedestrian Assignment Model (MPAM) was solved with an accuracy threshold of 10^3, ensuring convergence while maintaining tractability for the large network. The optimization process required approximately 8,100 MPAM evaluations, compared to over 1,048,000 evaluations that would be needed under full enumeration, representing a significant reduction in computational effort. The test network included roughly 1,800 OD pairs representing both work and leisure trips. The resulting Pareto front provided planners with a spectrum of trade-off solutions balancing travel cost, environmental quality, and construction cost, confirming that the NSGA-II approach can efficiently solve complex, behaviourally informed network design problems at a practical scale.

3 Comparing Results Across the Literature

Direct comparison of quantitative results across bicycle-network design studies is inherently difficult because the papers differ in four fundamental dimensions:

- **Problem formulation**. Studies do not optimise the same thing. Some grow new network edges [13, 19], others reallocate existing road space [14, 18], and some mix capacity reallocation with new links [27]. Growing edge approaches measure idealised structural improvements, whereas reallocation models reflect real-world constraints; their outcomes therefore quantify different phenomena.
- **Metric definitions**. Even when papers use the same terminology, underlying definitions differ. For

example, "directness" in Szell et al.[13] is the Euclidean–shortest-path ratio, whereas in Natera Orozco et al.[19] it compares bicycle paths to car paths. Although both are labelled "directness", the former measures geometric efficiency while the latter captures bicycle–car competitiveness.

- **Network size and morphology**. Case studies range from small districts with ∼100 edges [16] to large urban networks with over 14,000 edges [15]. Network size and street morphology strongly influence centrality, connectedness, and efficiency metrics, making their values non-comparable across cities.
- **Flow and demand assumptions**. Studies employ fundamentally different demand models: uniform OD matrices [13, 19], survey- or observation-based OD matrices [16, 22], or real cyclist trajectories [18]. Since network performance depends on routing behaviour and demand distribution, these choices significantly alter metric outcomes.

Taken together, these methodological differences mean that numerical performance measures across studies do not represent the same underlying constructs, and cross-paper numerical comparisons should therefore be interpreted qualitatively rather than quantitatively.

Comparison of Results Across Approaches The reviewed studies report a wide range of outcomes depending on the modelling approach adopted, with clear differences emerging between heuristic, MILP/LP and metaheuristic. Heuristic approaches such as Szell et al.'s growing-network model[13] show that simple rules like betweenness or closeness growth can rapidly consolidate cycling networks: the Largest Connected Component expands quickly, global efficiency rises to around 0.82 (approximately three times that of existing networks), and synthetic networks significantly outperform current real-world layouts. However, these benefits occur without behavioural realism, and car-network effects remain minimal except for a modest ∼5% increase in clustering under the fietsstraat assumption. A similar pattern appears in [19], where small, targeted missing-link additions increase Bogotá's cycling connectedness from 15% to 56% and improve bicycle-to-car directness from 6% to 48%, though bicycles remain roughly one-third less direct than vehicles.

In contrast, linear and mixed-integer programming models provide more coordinated and globally optimised interventions. Wiedemann et al.'s LP framework [14] demonstrates that reallocating lanes can decrease perceived bicycle travel times by over 35% while limiting car travel-time increases to under 20%. Integer relaxation introduces only a ∼5% optimality loss and enables computation on networks with up to ∼10,000 edges when parallelised. Structurally, when rebuilding the whole city from scratch the method converts roughly 60% of streets into one-way operations and reassigns about one-third of motorised lanes to cycling. Other MILP studies, such as Steinacker et al.'s long-term cycle-superhighway planning [27], find that both direct optimisation and backward percolation yield similar final network sizes (1300–1900 km), but direct optimisation achieves 8–10% higher welfare (NPV) by year 30. This gain comes with higher sensitivity to parameter uncertainty, whereas backward percolation is more robust but yields lower welfare early on. Trajectory-based MILPs (BL-AC and BL-GU) [18] further reveal trade-offs between coverage and continuity: BL-AC installs more segments but results in weaker continuous corridors, while BL-GU produces up to 49-segment high-continuity paths but covers fewer streets. Incorporating route choice spreads infrastructure geographically rather than producing a single dominant backbone, illustrating the behavioural impacts of infrastructure allocation.

The metaheuristic algorithm evaluated in Rastbin et al. [16] assesses its performance using a welfare-gain metric, defined as the improvement of each objective relative to the current network, normalised by the best value that objective could achieve if optimised in isolation. Using this measure, the model achieves an 84.24% welfare gain for F1 (total travel cost), indicating a substantial reduction in pedestrian travel effort. The environmental-quality objectives (F4)—including security, safety, walkability, sociability, sense of richness, and permeability—show similarly strong gains, ranging from 78% to 91%, demonstrating consistent enhancements across all qualitative dimensions of the pedestrian environment. Taken together, the average welfare gain of roughly 84% highlights the effectiveness of multi-objective optimisation in producing balanced improvements across conflicting criteria. Although the study focuses on pedestrians, the underlying insight—that carefully structured multi-objective methods can yield large, well-distributed network improvements—is directly relevant to cyclist-oriented infrastructure design.

3.1 Conclusion

The reviewed studies demonstrate that bicycle network design can be approached through a range of methodological frameworks. Together, these methodological perspectives reveal a landscape of complementary tools rather than a single dominant paradigm. However, a persistent challenge lies in integrating behavioural realism and computational scalability within a unified optimisation–evaluation framework. Addressing this gap motivates the present work, which aims to develop a modelling tool that combines scalable network optimisation with behaviourally informed evaluation to support evidence-based planning of cycling infrastructure in Malta.

4 Methodology

4.1 Network Representation

A critical step in the modelling pipeline is selecting an appropriate graph representation for the transport network. As outlined in section 2, the literature employs three main structures: undirected graphs, directed graphs, and directed multigraphs. The objective of this project is to propose a practical and implementable cycling network for Malta that improves bicycle connectivity while minimising disruption to the existing car network. For this reason, the transport network was modelled as a directed graph, which aligns with the majority of optimisation-based studies and more accurately reflects real-world travel patterns by capturing one-way streets and direction-specific attributes essential for OD-based flow assignment.

An undirected representation was deemed unsuitable, as it cannot encode directional restrictions or asymmetries in travel behaviour, both of which are common in Malta's dense urban street layouts. Directed multigraphs were also considered but ultimately not adopted. While they allow lane-level modelling of capacity and flow, this level of granularity exceeds the scope of the present study. Additionally, modelling lane-by-lane flow would introduce substantial computational overhead as the number of edges in the graph substantially increase, especially on the scale considered of all of Malta. A directed graph therefore provides the appropriate balance between behavioural realism, data availability, and computational tractability.

A directed representation was therefore adopted for routing, connectivity analysis, and OD-based path assignment. However, this did not imply that all edge attributes in the directed graph were treated as authoritative physical descriptions of the road. In particular, lane counts from OpenStreetMap were not interpreted directly at the level of directed arcs. Instead, the directed graph was used to represent directional accessibility and movement constraints, while lane supply was treated separately as a property of the underlying physical road segment. This distinction was necessary because the routing problem is inherently directional, but the corresponding lane metadata are not always direction-specific in the source data.

4.2 Feasibility Constraints for Lane Reallocation

Another critical step is selecting the problem formulation most appropriate for the Maltese context. As discussed in the literature, bicycle network design can be approached in three main ways: (i) creating new links between nodes (i.e., building new roads), (ii) reallocating space within the existing road network, or (iii) a hybrid of both strategies. For this project, the reallocation approach was chosen. This decision is motivated by Malta's severe spatial constraints. As a densely built island with limited remaining open space, expanding the road network by constructing new corridors is neither practical nor desirable, particularly given the importance of preserving the remaining green areas and avoiding further overdevelopment. A reallocation formulation therefore aligns with environmental, spatial, and policy considerations while constraining the search space to the existing network. Computationally, this avoids the need to generate and evaluate new edges, making the optimisation problem more tractable and better grounded in realistic planning constraints.

Most reviewed studies do not treat the existing cycling network as a constraint or starting condition. Reallocation-focused models such as Wiedemann et al.[14] and Liu et al.[18] operate on the full street network and implicitly assume that existing bike lanes can be redesigned. Generative approaches like Szell et al.[13] and Steinacker et al.[27] start from either an empty or fully developed network and entirely ignore existing infrastructure. Only Natera Orozco et al. [19]'s data-driven analyses, use the existing bike network explicitly, and even then it is employed primarily as a performance baseline rather than an optimisation input.

However, restricting interventions to existing links also reduces design flexibility. Without the ability to introduce new connections, the model cannot overcome structural bottlenecks inherent in the current street layout. This limitation is particularly evident in areas of Malta where residential grids consist entirely of single-lane, one-way streets (**??**, circle A). In such configurations, it is mathematically impossible to reallocate lanes to cycling without fragmenting the car network, because the street geometry provides no redundant capacity or bidirectional alternatives. Maintaining strong connectivity under reallocation requires corridors with at least two lanes or paired opposing one-way streets.

This issue is not unique to the present study: even advanced optimisation frameworks struggle under the same constraint. Wiedemann et al.'s LP formulation [14] explicitly notes that their algorithm is predisposed to reassign *one lane of any double-lane road* to cycling while preserving feasibility for cars. Their method guarantees strong connectivity only because the capacity constraint is bounded by the number of lanes; when streets have a single lane, the model cannot reallocate space without violating car reachability. In other words, their formulation also relies on the presence of multi-lane corridors and would fail in a network dominated by one-lane streets—precisely the condition found across many Maltese localities.

Similar challenges arise in regions such as circles B and C in **??**. Although a few bidirectional edges exist, many critical corridors still contain only one outgoing lane and function as articulation edges; reallocating them would break reachability and disconnect parts of the car network.

While some streets could technically be reallocated for cycling, constructing a continuous and fully segregated protected bicycle network is often infeasible under these structural constraints. To address this, the project adopts a *fietsstraat* (bicycle street) approach, similar to [13], for segments where lane reallocation would compromise car connectivity. In fietsstraten, cyclists and cars share the roadway, but cyclists receive priority and speed limits are reduced, improving safety while preserving the strong connectivity of the car network.

An additional strategy that was considered was to parametrise the model to increase the effective road capacity of single-lane streets by reallocating space currently used for on-street parking. In principle, removing parking could free sufficient road width to accommodate an additional traffic lane or a protected cycling facility. However, implementing such an approach was not feasible because the necessary data are not publicly available. OSM does not consistently provide road-width measurements or detailed information on the presence and extent of on-street parking along each segment. Without reliable width or parking data, any capacity expansion would require speculative assumptions, undermining the validity of the optimisation results. For this reason, the model focuses solely on reallocation of existing lane capacity rather than inferring latent space from parking removal.

4.3 Preprocessing

4.3.1 Network Extraction & Information Representation

Network Extraction The underlying street network of Malta was obtained through the publicly available OpenStreetMap (OSM) API. When loading the network, the parameter `simplify=True` was used. This setting simplifies the graph topology by removing interstitial nodes that only encode the geometry of curved street segments, while preserving the full original shape of each segment as a `geometry` attribute on the resulting edge. Consequently, nodes correspond more closely to true intersections and dead-ends, and edges correspond more closely to actual street segments rather than chains of short digitised fragments. This substantially reduces the number of nodes and edges in the graph, improving both memory efficiency and runtime for downstream analyses.

Simplification is also important for the validity of topological measures. In raw OSM-derived graphs, curved roads are often represented by multiple small edges connected through closely spaced geometry vertices. If left unsimplified, shortest paths that traverse a single physical street segment are distributed across several adjacent edges, which can fragment and distort edge-level centrality measures such as `edge_betweenness_centrality`. By collapsing these chains into a single edge between meaningful endpoints, the resulting centrality values better reflect the role of whole street segments in the network rather than artefacts of geometric digitisation [28].

However, if a simplified edge corresponds to multiple underlying OSM ways, their distinct attributes are retained as lists. While this aggregation is useful for reducing graph complexity, it introduces irregularities in attributes that would ordinarily be expected to take a single value. For example, two merged edges may carry highway tags `"residential"` and `"primary"`, resulting in a simplified edge with `highway=["residential","primary"]`. A single classification is required for downstream processing, particularly for safety-related evaluations. To resolve such conflicts, the least safe (i.e., highest-risk) tag was selected as the final attribute. This conservative choice reflects a cyclist-centred perspective: starting from the worst-case classification ensures that subsequent infrastructure allocation prioritises safety.

The parameter `retain_all=True` was also enabled to ensure that the entire OSM network is returned, including any disconnected subgraphs. This is important because the protected cycling infrastructure in Malta is highly fragmented, and discarding disconnected components at this stage would remove most of the existing bicycle network.

Modal Subnetwork & Master Graph Construction Two overlapping but distinct subnetworks were then extracted from the OSM data: the bikeable network, denoted as G_b, and the drivable network, denoted as G_c. Each edge in these subnetworks was annotated with Boolean attributes `bike_allowed` and `car_allowed`, indicating whether the edge is traversable by the corresponding mode. These attributes allow consistent modal tracking throughout the processing pipeline.

The two subnetworks were subsequently merged into a single composite master graph, G, which serves as the unified source of truth for all later stages of analysis. This integration is necessary because several edges are shared between the car and bicycle networks, and maintaining a single combined graph avoids inconsistencies when reallocating lanes or updating network attributes. After composition, only the largest weakly connected component of G was retained, as the project focuses solely on lane reallocation rather than constructing new roads. Consequently, any node unreachable in the composed graph would remain unreachable in all future network configurations.

Segment-level lane inventory and arc-level routing Although the analysis requires a directed graph, OpenStreetMap lane tags cannot be assumed to describe each directed arc independently. In OSMnx, when a road is not one-way, the path is added in its original direction and then added again in reverse, with the same way-level attribute dictionary attached to both directed edge additions. As a result, a tag such as lanes=1 may appear on both directed arcs of a two-way road, even though this should not be interpreted as one lane in each direction. Rather, it reflects the fact that the original OSM way carried a single lanes value and that OSMnx copied that value onto both directed representations. Conversely, if a two-way road has no lane tag in OSM, OSMnx may still generate arcs in both directions but without any lane attribute. Therefore, the directed graph is appropriate for routing, but not as the sole authoritative source for physical lane supply.

To address this, lane availability was represented at the level of the underlying physical segment rather than the directed arc. A temporary undirected representation was constructed to derive an authoritative segment inventory, where each segment stores the total number of available car lanes together with its geometry and road classification. This inventory was then linked back to the directed graph by mapping each directed arc to exactly one undirected segment. In implementation terms, the segment inventory is the authoritative representation of physical lane supply, whereas the directed graph remains authoritative for directional access and routing behaviour. This separation prevents the same physical lane supply from being mistakenly counted twice simply because a two-way street appears as two traversable arcs in the routing graph.

This design is also consistent with OSMnx's undirected conversion behaviour. The `to_undirected` conversion preserves parallel edges only when their geometries differ, which is useful here because opposite directions of the same carriageway can be collapsed into one authoritative segment, while separately mapped carriageways of a divided road remain distinct when their geometries are different. More generally, OSMnx notes that undirected street representations prevent double-counting the two directions of a two-way street, while still potentially treating divided roads as separate street entities. For the present study, this is desirable: a one-carriageway two-way street should contribute one shared pool of lane supply, whereas physically separate carriageways should be able to behave independently during reallocation.

Once segment-level lane supply had been established, all lane-reallocation operations were applied to this segment inventory first and only then propagated back to the directed graph. Consequently, changes such as converting a two-way street into a one-lane, one-direction car segment with a protected bicycle lane were implemented by updating the remaining segment-level lane supply and then modifying the accessibility of the corresponding directed arcs. In this way, the car_lanes, car_allowed, and reallocatable attributes on directed edges should be understood as derived operational state variables rather than as the primary physical truth of the road. This ensures that optimisation and routing still operate on a directed network, while infrastructure decisions remain grounded in a single non-duplicated representation of road space.

However, `OSMnx` provides and expects multi digraphs, typically these 'parallel' connections in the same direction are roads that have a physical barrier between them. These cases are rare (<1% in the case of Malta) and the lost of just a bit of information is worth for a more standardised approach. This is primarily done to have a standardised representation of the network. The multi digraph is converted to a digraph while retaining as much information as possible. In the case where two edges in the same direction between nodes occur these must be aggregated into a single edge. For numerical attributes such as length and grade an average value is taken to best

NOTE that all roads on the same edge should be the same type. I.E you cant have a freistatt and 'normal' car road on the same edge. And it doest make sense to have different speed limits on the same edge.

4.3.2 Attribute Enrichment

Graph Projection Before further preprocessing, the network was projected from geographic coordinates (latitude–longitude) to a planar coordinate reference system (CRS). Projection is essential because shortest paths and distances cannot be computed accurately using unprojected spherical coordinates, where units are expressed in degrees rather than metres. OSMnx automatically selects a suitable projected CRS for the study area based on its geographic location, choosing the local UTM zone that minimises distortion. This ensures that all spatial operations—such as distance calculations and edge lengths are carried out with consistent metric accuracy across the entire Maltese network.

Elevation and Grade Once the master graph was obtained, it was preprocessed and standardised to ensure that all required attributes were present. Elevation data for each node was retrieved using the Open Topo Data API, which provides values from the European Digital Elevation Model (EU-DEM) at a 25 m resolution. This was the finest publicly available elevation dataset for Malta, and its resolution is sufficient for urban-scale cycling analysis given the island's relatively gentle topography and absence of steep mountainous terrain.

However, not all nodes returned elevation values, so missing entries were imputed by averaging the elevations of their neighbouring nodes. This interpolation was performed iteratively because some nodes had neighbours that also lacked elevation data and required values to be propagated, until no further updates were possible, effectively producing a local linear interpolation. This neighbourhood-based approach was preferred over using a global mean, as elevation varies spatially and preserving local gradients is essential for computing realistic slope values. With elevation assigned to all nodes, the grade (steepness) of each edge was then calculated using the standard rise-over-run formulation. During this process, a small number of edges were found to have implausible gradient values, with absolute grades exceeding 100% (i.e., slopes steeper than $45°$), which are infeasible for road infrastructure and likely resulted from residual elevation artefacts. To ensure physical realism and prevent these outliers from skewing downstream analyses, grades were clamped to the range $\pm 20\%$ (approximately $11°$). This upper bound remains challenging for cyclists but represents a more realistic maximum slope found within Malta's road network.

Road-Class Consolidation Another preprocessing step involved merging semantically equivalent road classifications. For example, OSM distinguishes between "motorway" and "motorway_link", where the latter refers to short connector segments between major roads. For the purposes of this study, such distinctions are not meaningful, as both road types present a similar level of exposure and danger to cyclists. These categories were therefore consolidated to ensure consistent treatment during network analysis and lane reallocation.

Safety Classification Another preprocessing step involved classifying the safety level of each edge. Road segments where cyclists travel alongside motor traffic were labelled as cautious, reflecting the elevated exposure to vehicles. Primary roads and trunk roads were classified as dangerous, as these typically feature high traffic volumes, multiple lanes, and higher speeds. Additionally tunnels are also considered unsafe due to reduced visibility. Conversely, edges that are fully dedicated to bicycle use—such as segregated cycle paths or shared-use paths without vehicular access—were labelled as safe.

Speed limits Road segments are associated with different speed limits. However, in the Maltese Open Street Map (OSM) data, roads are generally not tagged with explicit speed limit information. As a result, speed limits had to be inferred.

According to *The Malta Road Code* [29], the maximum speed for private cars is $50\,\text{km/h}$ on urban roads and $80\,\text{km/h}$ on non-urban roads. These values represent upper bounds rather than uniform limits, since locally posted and variable speed limit signs may impose lower limits on specific roads. For example, some built-up areas are subject to a $30\,\text{km/h}$ limit [29]. To approximate this variation while remaining consistent with the national upper bounds, speed limits were assigned based on road type, as shown in table 1.

This was implemented using the `highway` classification, under the assumption that the intended function of a road is broadly correlated with its speed limit. For example, trunk roads can reasonably be expected to have higher speed limits than residential streets. Although this approach does not reproduce the exact legal speed limit of every road segment, it provides a practical approximation in the absence of complete tagged data.

Road type	Speed limit (km/h)
Trunk	70
Primary	60
Secondary	60
Tertiary	50
Residential	30
Service	30

Table 1: *Assigned speed limits by road type*

These assigned values were selected to reflect the relative hierarchy of road classes while remaining within the maximum limits established by the Maltese Road Code.

Reallocatability Finally, each edge was assigned a `reallocatable` Boolean attribute. An edge was marked as non-reallocatable if it did not belong to the car network or if bicycles were not permitted on that segment (i.e., it does not appear in the bikeable network and is classified as dangerous). All remaining edges were considered reallocatable.

4.3.3 Summary

In summary, after the preprocessing pipeline the final network is represented as a directed graph whose edges contain all attributes required for subsequent optimisation. Each edge stores information obtained from OSMnx and Open Topo Data, including length (in metres), the number of car and bicycle lanes, and node elevations. From these, additional attributes are derived, such as edge grade, safety classification, and whether the segment is eligible for reallocation by the optimisation algorithm.

4.4 Synthetic Travel Demand and OD Matrix Construction

This subsection describes how travel demand is represented in the model and how the synthetic trip-generation process is adapted to the observed travel patterns of

Malta. As discussed previously in 2.1.2, there are two principal ways of representing travel demand. The first is through trajectory data, which record the exact routes taken by individuals between origins and destinations. However, such data are typically not publicly available and are not available for Malta. Instead, an OD matrix is used, which specifies only the number of trips originating in one area and ending in another.

To construct a generalisable demand model, an algorithm is developed to generate synthetic trip-level origin–destination pairs that are consistent with observed aggregate travel patterns. Rather than directly assigning observed trips to fine spatial units, the model combines empirical regional travel distributions with synthetic within-region destination choice. In particular, the observed Malta data from [30] are available only at the regional level, whereas the model requires origins and destinations at the locality or polygon level. The observed regional OD structure is therefore used directly to constrain destination-region choice, while the finer spatial distribution of trips within regions is generated endogenously. This allows the framework to preserve the aggregate regional structure seen in the data while remaining transferable to other settings where only coarse-grained demand information is available.

4.4.1 Observed Malta Regional Travel Patterns and Regional Harmonisation

The empirical travel-demand inputs are taken from the *National Household Travel Survey 2021* [30]. These data provide aggregate information on trips between Maltese regions, as well as the distribution of outward trip purposes by destination region. Together, these datasets provide the empirical regional structure that guides the synthetic trip-generation process.

These regional data provide an important empirical benchmark, but their aggregate nature also imposes clear limitations. Since the data are defined only at the regional level, they do not identify the specific localities or individual origins and destinations involved in each trip, nor do they provide route-level information. They are therefore suitable for constraining broad inter-regional demand structure, but not for directly reproducing locality-level flows.

An unexpected mismatch was identified between the locality–region assignments defined in OSM and those used in the OD matrix. OSM follows administrative boundaries [31], whereas the NSO adopts the NUTS classification defined by Eurostat [32]. For example, Ħ'Attard falls within the Northern Region under the administrative boundary system, but is classified as part of the Western Region under the NUTS system. As a result, the locality–region mapping could not be extracted directly from OSM. To ensure consistency with the OD matrix provided by the NSO, the mapping was instead defined manually according to the NSO regional classification.

4.4.2 Regional Destination Choice and Trip Purposes

In the synthetic demand model, travellers may undertake eight outward trip purposes: commuting, education, escort education, shopping, personal errands, medical, recreation, visiting, and other. Each trip purpose is associated with a relevant subset of destination types, which defines the candidate destination set D_p. For example, shopping trips may be assigned to destinations such as `mall`, `marketplace`, or `shop`, while commuting trips may be assigned to destinations such as `bank`, `office`, or `shop`. As these examples illustrate, different trip purposes may map to overlapping destination categories.

In the implemented model, trip generation is structured hierarchically. First, a destination region is sampled conditional on the traveller's origin region, using the observed inter-regional trip matrix from the NSO. The observed region-to-region trip counts are row-normalised so that each row sums to one, yielding the conditional distribution $P(r_d \mid r_o)$. This captures the broad spatial structure of travel demand at the regional level. Second, conditional on that chosen destination region, a trip purpose is sampled using the observed outward purpose shares associated with that destination region. These purpose counts are likewise normalised within each destination region so that they form the conditional distribution $P(p \mid r_d)$. The interpretation is therefore not that trip purpose depends directly on where the traveller lives, but rather that the distribution of purposes varies according to the type of region to which the traveller is travelling. Both the regional destination-choice matrix and the purpose-by-destination-region matrix are therefore required: the former determines *where* trips are directed at the aggregate regional level, while the latter determines *why* those trips are made once a destination region has been selected.

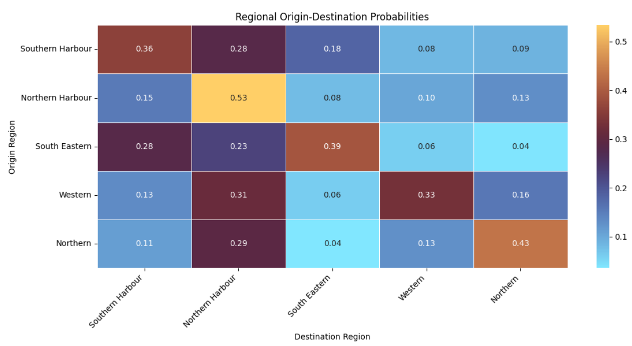

Figure 2: *Destination-region shares conditional on origin region*

The destination-region matrix (fig. 2) shows that intra-regional travel is generally the most common outcome, with the diagonal entries being the largest for all five regions. This is especially pronounced for Northern Harbour, where more than half of trips remain within the region, and for Northern, where intra-regional travel also accounts for a large share. At the same time, the matrix highlights several strong inter-regional links. In particular, Northern Harbour acts as a major destination for trips from all other regions, indicating its importance as a national activity centre. Southern Harbour also attracts substantial flows, especially from Southern Harbour itself and from South Eastern. Overall, the matrix suggests that travel demand in Malta is structured around both strong local containment within regions and strong attraction toward the harbour regions, particularly Northern Harbour.

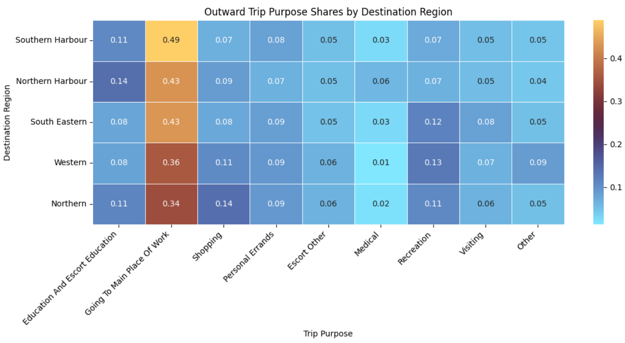

Figure 3: *Outward trip-purpose shares by destination region*

The purpose-by-destination-region matrix (fig. 3) shows that across all regions the dominant outward purpose is going to the main place of work, accounting for roughly one third to one half of trips. Southern Harbour has the highest work-related share, at just under one half of outward trips, while Northern and Western display somewhat more diversified purpose profiles. Education and escort education, shopping, and recreation form the next most important categories, though their magnitudes remain well below work-related demand. In particular, Northern has the highest shopping share, while Western and South Eastern display comparatively stronger recreation shares. Overall, the matrix suggests that regional variation in outward travel is driven primarily by the concentration of employment and, to a lesser extent, by differences in secondary activity types.

4.4.3 Population Data and Origin Weights

Population data are used to determine the relative strength of trip production across residential areas. Population counts at the locality level are taken from the *Census of Population and Housing 2021* [33]. Using census-based population weights is more realistic than relying solely on proxy measures such as the surface area of residential land, since area-based methods may overestimate demand in low-density suburban areas or underestimate it in compact high-density urban areas.

A limitation of the available population data is that they are reported at the locality level rather than for individual residential polygons. Consequently, when a locality contains multiple separate residential areas, its population cannot be assigned directly to each one. To address this, the locality's total population is distributed across its residential areas in proportion to their surface area, providing an approximate weighting for trip generation. Thus, population data determine trip-production weights at the locality level, while residential land area is used only to apportion those weights among multiple residential areas within the same locality.

4.4.4 Trip Generation Algorithm

Travel demand cannot be represented as purely random, since people do not travel arbitrarily but instead follow broad and recurring patterns. Although it is difficult to model the exact behaviour of each individual, statistical aggregation makes it possible to infer these general trends. For this reason, destination choice is modelled using a probabilistic gravity-based formulation, similar to approaches used in previous transport-demand modelling studies [16].

The trip generation algorithm assumes that individuals primarily begin their journeys in residential areas. For each sampled trip, an origin residential polygon is first selected with probability proportional to its assigned population weight. Let o denote the sampled origin.

Given the origin region r_o, a destination region r_d is then sampled according to the empirical conditional distribution

$$P(r_d \mid r_o),$$

derived from the observed regional OD matrix. After this, a trip purpose p is sampled according to the observed purpose shares conditional on the selected destination region,

$$P(p \mid r_d).$$

Once r_d and p have been selected, the available destinations are restricted to the subset of point of interests (POIs) within region r_d whose type is compatible with purpose p. Denoting this set by D_{p,r_d}, the probability of selecting destination $j \in D_{p,r_d}$ is given by

$$P(j \mid o, p, r_d) = \frac{e^{-\beta d(o,j)}}{\sum\limits_{k \in D_{p,r_d}} e^{-\beta d(o,k)}}. \tag{1}$$

Here, $d(o,j)$ is the Euclidean distance between origin o and destination j, and β is the distance-decay parameter.

Larger values of β impose a stronger penalty on distant destinations, thereby increasing the likelihood of shorter trips. The value of β is calibrated, as discussed in the following subsection.

To construct the sets of admissible origins and destinations, residential areas and relevant POIs were extracted from OSM. The extracted destination opportunities included features derived from `office`, `shop`, and `amenity`. Since the `amenity` category contains a wide variety of subtypes, it was further disaggregated so that only relevant categories were retained explicitly, while the remainder were grouped under `other`. In addition, residential centroids were also included as destination opportunities for trip purposes such as visiting.

The destination-selection algorithm therefore operates as follows. First, an origin is sampled from the set of residential polygons, with probabilities weighted by the population assigned to each polygon. Second, a destination region is selected probabilistically using the observed regional destination shares for the origin region. Third, a trip purpose is selected probabilistically using the observed purpose shares for that destination region. Fourth, the available destinations are filtered according to both the selected region and the destination types associated with the selected purpose, thereby forming the set D_{p,r_d}. Finally, the gravity-based probability model is applied to select the specific destination.

In the survey data, return-home travel appears as an explicit trip purpose. However, this category is not modelled as an independently sampled purpose in the synthetic demand generator. Instead, the current implementation adopts a simplified home-based tour structure: an outward trip is first generated from a residential origin to a sampled destination, and this is then followed by an explicit return trip from that destination back to the original home location.

As a result, the final synthetic demand consists of paired outward and return-home trips rather than isolated one-way journeys. This is important because assignment is performed on a directed road network, where the outbound and return legs may use different links or directions of travel. Modelling only the outward leg would therefore bias the demand representation by loading traffic in one direction only, whereas including the return leg ensures that both directions of the home-based journey are represented.

This choice was made to retain the essential home-based structure of daily travel while avoiding the additional complexity of modelling full activity chains with multiple intermediate stops and sequential destination choices. Although a full travel-chain model would provide a more behaviourally detailed representation of daily mobility, implementing such a framework would require additional assumptions and behavioural structure that exceeded the scope of the present study.

4.4.5 Calibration

The distance-decay parameter β controls the strength with which distance discourages destination choice in the gravity-based component of the synthetic trip generator. Small values of β imply weak distance deterrence, meaning that destinations within the selected region and purpose category are chosen almost independently of their distance from the origin. Conversely, large values of β strongly penalise distant destinations and therefore generate shorter, more localised trips.

Calibration was performed by comparing the mean shortest-path length of the synthetic trips against the target average trip distance for Malta, taken to be 6.1 km [34]. For each candidate value of β, a synthetic OD sample was generated, the resulting origin–destination pairs were assigned to the road network using shortest paths, and the weighted mean path length was computed. The calibration error was defined as

$$E(\beta) = \left|\bar{d}_{\text{synthetic}}(\beta) - \bar{d}_{\text{target}}\right|, \tag{2}$$

where $\bar{d}_{\text{synthetic}}(\beta)$ is the mean network path length produced by the synthetic demand model and $\bar{d}_{\text{target}}$ is the target average trip distance.

A logarithmic sweep was used because β appears inside an exponential distance-decay term. Equal changes in the order of magnitude of β therefore have a more meaningful behavioural interpretation than equal linear increments. The broad sweep tested values between 10^{-6} and 10^{-2}, allowing both weak and strong distance-deterrence regimes to be evaluated.

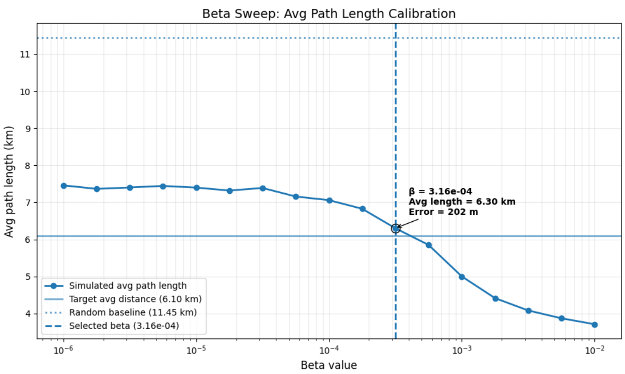

Figure 4: Calibration of the gravity-model distance-decay parameter β.

The results in fig. 4 show that very small values of β produce trips that are too long, with average path lengths of approximately 7.3–7.5 km. In this range, the distance penalty is too weak to substantially affect destination choice. As β increases, the average trip length decreases, indicating that the gravity component increasingly favours closer destinations. At very high values of β, the model produces trips that are too short, suggesting excessive localisation of destination choice.

The best value in the broad sweep was $\beta = 3.16 \times 10^{-4}$, which produced an average path length of 6.30 km. This corresponds to an absolute error of 202 m relative to the 6.10 km target. The calibrated value therefore reproduces the target average trip length with reasonable accuracy, with an error of approximately 3.3%. Since the synthetic OD sample is subsequently used to estimate relative edge importance rather than to reproduce exact traffic counts, this level of agreement is considered sufficient for the purposes of the model.

As a simple benchmark, the calibrated model was compared with a purely random OD generator in which origins and destinations are sampled from network nodes subject only to a minimum Euclidean separation constraint. This random baseline produced an average path length of 11.45 km, corresponding to an error of approximately 5.35 km. The baseline is not intended to represent a realistic behavioural model, since it does not preserve the empirical regional OD structure, population-weighted origins, or purpose-specific destination constraints. Rather, it provides a sanity check showing that unconstrained random OD assignment produces unrealistically long trips for Malta and is unsuitable as a basis for demand representation.

It is important to note that only the spatial distance-decay parameter was calibrated. The regional destination shares $P(r_d \mid r_o)$ and purpose shares $P(p \mid r_d)$ were applied directly from the available survey data. Calibration therefore adjusts the latent within-region spatial destination-choice mechanism while preserving the observed aggregate regional travel structure.

The calibration results also show that the empirical regional constraints alone are not sufficient to reproduce the observed average trip length. Even when origins are population-weighted and destination regions are sampled from the observed regional OD matrix, very small values of β still produce average path lengths of approximately 7.3–7.5 km. In this regime, the model preserves the regional travel structure and purpose-specific destination filtering, but the final destination choice within the selected region is almost insensitive to distance. The resulting trips remain too long relative to the 6.1 km target. This indicates that population data, regional OD shares, and destination availability are important, but that they must be combined with a gravity-based distance-decay mechanism to generate realistic trip lengths. The calibrated value of β therefore captures an essential behavioural component of destination choice that is not explained by regional restrictions alone.

4.5 Optimisation Algorithms

To evaluate which optimisation strategy produces the most effective cycling network for Malta, three classes of segment-selection algorithms were implemented: heuristic, mathematical, and metaheuristic approaches.

Objective 1 - Heuristic The heuristic methods follow directly from the techniques reviewed in section 2.3.1. Centrality-based heuristics were implemented using edge betweenness and closeness centrality, prioritising segments that are topologically critical or spatially central [13]. In addition, the component-based heuristics of [19] were replicated, including the L2S, L2C, R2C, and CC strategies. These heuristics aim to consolidate fragmented cycling infrastructure by reconnecting components in a manner that balances cohesion, spatial efficiency, and investment cost.

Although some studies [13, 14, 27] explore backward (bike→car) strategies to avoid path-dependence and analyse growth sequences, these methods are primarily valuable for heuristic benchmarking rather than practical planning. In this project, the primary goal is to design a forward-looking cycling network for Malta, starting from the current car-dominated layout. Since LP/MILP approaches already provide globally coordinated allocations without relying on sequential growth, implementing backward strategies for these optimisation models would add substantial complexity with little analytical benefit. Similarly Wiedemann et al. [14] only implemented backward heuristic strategies for better comparability of models.

Objective 2 - Mathematical Approaches The second objective is to formulate the network design task using the more rigorous mathematical optimisation approaches discussed in section 2.3.3. This involves constructing a linear or mixed-integer linear programming model that computes a globally optimal solution subject to the constraints of the Maltese road network. Compared to the heuristic strategies, the mathematical formulation incorporates additional detail, including edge grade, mode-specific perceived travel times, and origin–destination travel demand between localities. The OD matrix used in this model will be derived from Saliba et al. [35], who inferred travel flows for Malta using anonymised Call Detail Record (CDR) data from GO.

Objective 3 - Metaheuristic The third objective is to implement a metaheuristic optimisation framework capable of exploring the solution space more flexibly than exact mathematical programming. Building on the approaches discussed in section 2.3.4, a multi-objective evolutionary algorithm, NSGA-II will be applied to balance competing criteria—including cycling accessibility, car travel performance, and network safety—without requiring the strict linearity or relaxations imposed by LP/MILP models. The metaheuristic formulation evaluates candidate network configurations using the same enriched attribute set (grade, perceived travel times, and OD-based flows) but searches the design space stochastically through mechanisms such as mutation, crossover, and non-dominated sorting. This allows the identification of a diverse Pareto-optimal set of solutions, enabling planners to assess trade-offs across different policy priorities rather than converging to a single prescriptive optimum.

4.5.1 Evaluation Metrics

At each iteration, when the optimisation model proposes a new segment for inclusion in the cycling network, the resulting street layout is evaluated using four groups of metrics. These groups distinguish between the structure of the protected cycling network, the quality of routes available on the full bikeable network, the system-level trade-off between cycling improvement and car-network degradation, and the spatial reach of the resulting network.

Protected cycling network structure. The first group of metrics evaluates the topology of the safe, protected bicycle subnetwork. Since protected infrastructure is the part of the network intended to provide a high-quality cycling environment, these metrics are computed on the protected cycling graph rather than on the full bikeable network. Where component-level metrics are required, the largest connected component (LCC) is defined by total edge length rather than node count, as length better reflects meaningful network extent, particularly when intersection density varies across areas.

Connectedness is assessed through two measures: the number of connected components in the protected network and the total length of its LCC. The number of components captures fragmentation, while LCC length measures the extent of the largest continuous, navigable protected cycling corridor. Together, these metrics indicate whether proposed interventions consolidate isolated fragments into a more coherent cycling network.

Centrality is evaluated across the LCC to describe the internal structure of the protected network. Betweenness centrality captures the extent to which edges act as important shortest-path corridors, while closeness centrality reflects accessibility within the component. Higher centrality values indicate that the protected network supports more efficient movement through its structure, although these measures should be interpreted alongside connectedness because centrality is only meaningful once a sufficiently large connected component has formed.

Route directness on the full bikeable network. Directness evaluates how competitive cycling routes are relative to car routes for the same OD demand. Unlike the connectedness and centrality metrics, directness is not computed only on the protected cycling subnetwork or on its LCC. Instead, it is calculated on the full bikeable network, which includes both protected cycling infrastructure and streets that remain available for cycling. This reflects the fact that cyclists may still use unprotected streets, but that these routes may be less attractive than protected alternatives.

Following the bicycle–car directness approach of Natera et al. [19], directness is defined as the ratio between the shortest car-route length and the shortest perceived bicycle-route cost for each OD pair. For an OD pair (o, d), this is given by

$$D_{od} = \frac{L_{od}^{car}}{L_{od}^{bike,perc}},$$

where L_{od}^{car} is the shortest-path length through the driveable network, and $L_{od}^{bike,perc}$ is the shortest perceived cycling cost through the full bikeable network. The reported directness value is the demand-weighted mean across all reachable OD pairs:

$$D = \frac{\sum_{(o,d)\in\Omega} w_{od} D_{od}}{\sum_{(o,d)\in\Omega} w_{od}}.$$

This differs from conventional Euclidean directness, where route efficiency is measured as the ratio between straight-line distance and network distance. Here, the metric instead compares cycling against car travel, making it a measure of modal competitiveness. Values closer to one indicate that cycling routes are perceived as similar in cost to equivalent car routes, while lower values indicate that cycling involves greater detour, discomfort, or exposure to less suitable infrastructure.

The use of perceived cycling length is important. A route may be physically short but unattractive if it requires cycling on unprotected or high-stress streets. Conversely, adding protected infrastructure can improve directness even when the geometric length of the route changes only slightly, because the perceived cost of cycling along that route is reduced. Directness therefore captures both spatial efficiency and route quality. It should not be interpreted purely as a geometric measure of straightness, but as an OD-weighted indicator of how direct, comfortable, and competitive cycling is within the full bikeable network.

Baseline-relative accessibility trade-off. A further pair of *vs baseline* metrics is used to quantify the trade-off between cycling improvement and car-network degradation. This follows the Pareto-style evaluation logic of

Wiedemann et al. [14], where improvements to the bicycle network are assessed relative to their impact on car accessibility. At iteration 0, before any additional segment is reallocated, the total OD-weighted perceived cycling cost and the total OD-weighted car cost are stored as baseline values, denoted by C_b^0 and C_c^0, respectively. At each subsequent iteration i, the updated costs C_b^i and C_c^i are recomputed over the same OD demand set.

The relative cycling gain is defined as

$$G_b^i = \frac{C_b^0 - C_b^i}{C_b^0},$$

where positive values indicate that the proposed cycling network reduces the total perceived cycling cost relative to the initial network. Conversely, the relative car harm is defined as

$$H_c^i = \frac{C_c^i - C_c^0}{C_c^0},$$

where positive values indicate that the intervention increases total car travel cost relative to the baseline. These two measures express the central planning trade-off: how much perceived cycling accessibility improves for a given deterioration in car accessibility.

Because interventions are applied cumulatively, both baseline-relative metrics are expected to be monotonically non-decreasing over the course of an optimisation run. As additional segments are converted into protected cycling infrastructure, perceived cycling cost should either decrease or remain unchanged, producing a non-decreasing cycling gain. Conversely, as road space is removed from the driveable network, car travel cost should either increase or remain unchanged, producing a non-decreasing car harm metric. In practice, this monotonicity provides a useful consistency check on the evaluation procedure: unexpected decreases may indicate changes in route eligibility, graph construction, or penalty calibration rather than a genuine improvement.

For car trips, some OD pairs may become unrouteable as the driveable network is progressively reduced. This can occur either because no directed path remains between the origin and destination, or because the origin or destination itself is no longer accessible from the car network after adjacent street space has been reallocated to protected cycling infrastructure. Rather than omitting these OD pairs from the evaluation, an unrouted-trip penalty is assigned. This ensures that lost car accessibility is counted as part of the aggregate car harm instead of disappearing from the metric. The penalty therefore represents the inconvenience imposed on drivers when a trip can no longer be completed by car within the modified network. It is chosen to be sufficiently large so that an unrouteable trip is treated as worse than a feasible detour, preserving the interpretation of unrouteability as additional car-network harm.

Spatial coverage. Finally, coverage is used to evaluate the spatial reach of the proposed cycling network. Following Szell et al. [13], coverage measures the proportion of the city that lies within a specified buffer distance, typically 500 m, of the bicycle network. This captures how accessible the infrastructure is to potential users and complements the topological indicators by assessing whether the network serves residential areas, workplaces, and key destinations.

Together, these metrics provide a quantitative framework for comparing network configurations across iterations, cities, and growth strategies. The protected-network metrics describe structural consolidation, directness measures perceived route competitiveness, the baseline-relative metrics quantify the bike–car trade-off, and coverage captures spatial accessibility.

4.5.2 Estimating Cycling Uptake

A key issue in this evaluation is that the proposed cycling network should not be tested under present-day modal shares alone. Doing so would risk systematically underestimating its performance, because Malta's current mode split reflects a network in which cycling is both marginal and structurally constrained. If the new infrastructure is assessed while holding today's travel preferences fixed, the model would implicitly assume no induced uptake of cycling, even though the intervention is specifically intended to remove barriers to cycling. This would bias the comparison against the proposed network by assuming that roughly the same volume of motor traffic must still be accommodated on a reallocated road space.

A more appropriate approach is therefore to simulate the candidate networks under a plausible future cycling uptake scenario. For Malta, the NHTS provides an empirical basis for such an assumption. As noted earlier, only 0.7% of all trips in Malta were made by bicycle in 2021 [30]. However, among respondents who did not use cycling for any trip, the survey also asked for the principal reason why they did not cycle more often. The most commonly reported barriers were inability to ride a bicycle (20%), road safety concerns (17%), and poor-quality cycling infrastructure (13%) [4]. These results suggest that a substantial share of suppressed cycling demand is linked not to trip impossibility, but to remediable external conditions, particularly safety and infrastructure quality. In which adding protected cycling lanes addresses these main issues.

This does not imply that the 17% and 13% figures can be translated directly into future cycling mode share. Rather, they indicate latent demand: a segment of the population for whom cycling may become a feasible option if the principal barriers are reduced. At the same time, the survey also indicates that not all non-cyclists are realistically convertible, since some respondents reported inability to ride, ill-health, age-related constraints, or simple lack of interest as their primary reason for not cycling. The aim, therefore, is not to project an upper-bound theoretical maximum, but to define a conservative and policy-relevant scenario for network testing.

On that basis, a target cycling mode share of 10% was adopted as an ambitious but plausible scenario for Malta. This sits slightly above a broader European benchmark. Special Eurobarometer 495 reports that 8% of EU28 respondents identified a privately owned bike or scooter as their main mode of transport on a typical day, compared with 2% in Malta [36]. This is not directly equivalent to bicycle mode share, and the Eurobarometer category also includes scooters. Nevertheless, it provides a useful comparative indication that two-wheeled personal mobility remains substantially less prevalent in Malta than in the wider European context. In that sense, a 10% cycling scenario can be seen as ambitious, but not implausible, under conditions of materially improved infrastructure.

4.5.3 SUMO

Objective 3 — Agent-Based Modelling After generating candidate cycling networks using the heuristic approaches, an agent-based model (ABM) was used to compare their likely behavioural performance in a more realistic setting. Whereas the optimisation methods rely primarily on topological and aggregate indicators—such as connectedness, centrality, directness, and mode-specific travel times—the ABM simulates traveller-level decisions, including route choice, perceived safety, and the potential substitution of car trips with bicycle trips. In this way, the ABM complements the earlier optimisation stages by evaluating not only how well a proposed network is structured, but also how it may actually be used in practice.

4.6 Traffic Simulation Software

Several traffic simulation platforms are widely used in transport modelling, including SUMO, MATSim, PTV Vissim, and Aimsun Next. SUMO is an open-source, microscopic, continuous, and multimodal traffic simulator designed for large networks. MATSim is an agent-based transport simulation framework in which travellers iteratively optimise their daily plans. PTV Vissim and Aimsun Next are commercial platforms used extensively in professional traffic studies, with both supporting microscopic simulation and Aimsun also supporting dynamic traffic assignment.

SUMO was selected for this work because it best matched the requirements of the study. The methodology developed in this dissertation generates origin–destination demand externally and assigns it to a road network derived from OpenStreetMap. SUMO is particularly well suited to this workflow because it is open source, supports microscopic simulation on large networks, and provides tooling for importing and processing OpenStreetMap-based scenarios. These characteristics made it more appropriate for this study than a full activity-based framework such as MATSim or proprietary software such as Vissim and Aimsun.

At a high level, SUMO simulates the movement of individual vehicles through a time-varying road network. The network is represented by junctions and road segments, while demand is provided as trips, flows, or routes. Vehicles are inserted into the network and move according to microscopic behavioural models and routing rules, interacting with other vehicles and traffic controls as the simulation progresses. As a result, traffic conditions such as delay and congestion arise from the simulated network interactions themselves.

4.7 SUMO Pipeline

The simulation pipeline used for SUMO required a different graph representation from that used in the earlier Python-based network analysis. In particular, SUMO requires an *unsimplified* road graph rather than the simplified graph typically used for network generation and evaluation. In an unsimplified graph, each collapsed edge segment is preserved explicitly, which is necessary because SUMO models traffic movement at the lane and junction level. By contrast, the simplified graph used earlier in the pipeline is more suitable for analysis and road-space reallocation, since consecutive road segments can be merged into single logical edges without losing the high-level topology needed for optimisation.

For the baseline network this distinction is relatively straightforward to handle, since the unsimplified graph can simply be exported before any simplification takes place. The proposed reallocated network, however, introduces an additional complication. The proposed design is generated and evaluated on the simplified graph, because simplification is needed to make the reallocation process tractable and to reason about road segments at a meaningful level. However, SUMO cannot directly consume this simplified representation. As a result, all modifications made to the simplified proposed graph must be propagated back onto the corresponding unsimplified graph before conversion to SUMO format. This propagation is performed through the merged-edge relationships recorded during simplification, allowing each simplified edge decision to be mapped back to the set of original edge segments from which it was formed.

This step is particularly important because it is not sufficient to treat the simplified and unsimplified graphs as interchangeable views of the same object. They are related, but they do not preserve a one-to-one edge correspondence. A single simplified edge may represent multiple underlying segments in the unsimplified graph, each of which must be updated consistently if the intended infrastructure change is to be reflected in simulation. This became one of the main challenges in integrating the Python modelling pipeline with SUMO, since the reallocation logic is naturally expressed over simplified edges, while the simulation logic ultimately depends on the full unsimplified network structure.

A separate implementation caveat concerns OSM XML export rather than the methodological representation of lane supply itself. OSMnx's all_oneway setting is documented as intended only for cases where the graph will subsequently be saved with save_graph_xml, since it forces ways to be added as one-way while preserving original node order. The documentation for save_graph_xml further states that the graph should be unsimplified, unprojected, and created with ox.settings.all_oneway=True for the function to behave properly. For this reason, all_oneway=True was treated as a requirement of the export pipeline used for SUMO conversion, rather than as part of the conceptual graph model used for optimisation and lane reallocation. In other words, the segment-level lane methodology described above remains the authoritative modelling choice, while the all-oneway configuration is only an implementation constraint required for reliable XML serialisation.

Once the updated unsimplified network is obtained, it can be converted into SUMO's native network format using the `netconvert` utility. The SUMO network format stores the road network as XML, with explicit representations of nodes, edges, lanes, and junction connections. This representation is considerably more detailed than the graph used earlier in the pipeline, because SUMO must encode both topological connectivity and the operational rules that govern vehicle movement through the network. These include lane-level access permissions, permitted turning movements, and junction geometry. For example, two lanes may continue in the same direction along an approach road, but only the left-most lane may be allowed to make a left turn at the next junction, while the adjacent lane is restricted to through movements. Such distinctions are essential in microsimulation, since they directly affect queue formation, lane choice, and route feasibility. Accordingly, `netconvert` acts as the stage that transforms a high-level road graph into a traffic network that can be executed meaningfully within SUMO's simulation model.

A further implication of this conversion step is that, although many source edge identifiers remain traceable in the SUMO network, the representation is no longer convenient to use as a direct continuation of the Python-side graph. SUMO introduces lane-level objects, explicit turning connections, and, where enabled, internal junction edges, so the operational network used in simulation is more elaborate than the graph on which the reallocation logic was originally applied. This makes a direct transfer of precomputed origin nodes, destination nodes, or full paths from the earlier pipeline possible only with additional custom processing around SUMO-specific network structures. Rather than building such a brittle interface, the workflow instead follows SUMO's native demand-generation approach based on traffic analysis zones. This is also consistent with established microscopic simulation practice, where OD demand is commonly grouped by TAZ and then assigned within the simulation network, as in the Zurich case study by Fulton et al. [26] , where OD travel times and route groupings were likewise defined over TAZs rather than exact point-to-point mappings.

For this reason, trip generation in SUMO was based on the more standard Traffic Assignment Zones (TAZ)-based approach rather than by directly supplying paths or even exact origin and destination nodes. In this approach, trips are defined between zones rather than between precise graph coordinates. For this work, each zone was defined at the locality level. This choice was methodologically appropriate because the demand data had already been aggregated spatially to localities. The approach avoids the brittle assumptions that would arise from trying to preserve exact node mappings across graph simplification, reallocation, and SUMO conversion, while still maintaining general model trends.

The locality polygons were exported and aligned to the SUMO network so that they could be used as district definitions. These district polygons were then processed using SUMO's `edgesInDistricts.py` utility, which extracts the network edges lying within each zone. This was performed separately for cars and for cyclists, so that each mode only received edges that are traversable for that vehicle class. This separation was necessary because the accessible network differs by mode, particularly after road-space reallocation where some links may remain valid for bicycles but not for general motor traffic, or vice versa. As a result, the specific edges available within each zone can vary by mode, even if the underlying zonal demand definition remains the same.

Demand was also modelled as time-dependent rather than static. The available OD data provided an hourly breakdown of trips, and this structure was preserved by defining separate OD matrices for each hourly interval. Using hourly intervals serves two purposes. First, it reflects the granularity of the source data directly, avoiding the need for arbitrary temporal smoothing. Second, it enables analysis of network behaviour under different demand intensities across the day, from relatively quiet periods to peak traffic conditions. This is important for evaluating the effect of the proposed reallocation under stress, since a network that performs adequately in off-peak conditions may still fail under rush-hour demand. The hourly formulation therefore allows a more granular assessment of throughput, congestion onset, and the extent to which the network can absorb increasing demand before jamming occurs.

With the zonal definitions and OD matrices in place, trips were generated using SUMO's `od2trips` tool. This tool samples specific departure and arrival locations from within the valid edges of each origin and destination zone. The use of `od2trips` means that the exact start and end points of individual trips are no longer identical to those produced in the earlier Python pipeline. However, this loss of point-level fidelity is acceptable in this context because the calibrated demand was already aggregated to the locality level. The important characteristic to preserve is therefore the spatial pattern of movement between localities, rather than the exact curb-level origin and destination of each traveller. While this approach does not guarantee equally realistic spawning locations in every possible locality definition, it provides a robust and SUMO-native way of generating trips that remains consistent with the available demand data and the converted simulation network.

After trip generation, routes were computed using SUMO's routing tools. In this workflow, the relevant choice was between `duarouter` and the more advanced `duaIterate` assignment procedure. `duarouter` is SUMO's standard tool for converting trip definitions into explicit route files on a given network, and it also supports repairing disconnected or imperfect demand definitions where necessary. By contrast, `duaIterate` performs repeated routing and simulation in order to approximate a dynamic user assignment, making it suitable when the objective is to model route adaptation over successive iterations rather than to construct a single consistent set of routes. In this study, `duarouter` was selected because the primary requirement was to generate valid and comparable route files for each scenario on the final converted SUMO network. An iterative assignment procedure was not required, since the objective was not to estimate an equilibrium response in route choice, but to compare baseline and reallocated networks under a common and reproducible routing workflow. TODO computionally expensive ig.

Overall, the SUMO pipeline can therefore be understood as a staged translation process. Reallocation decisions are first made on the simplified graph, then propagated back to the unsimplified graph, converted into SUMO's XML-based network representation, linked to locality-based traffic analysis zones, combined with hourly OD demand, transformed into trips through zonal sampling, and finally routed using `duarouter`. This design sacrifices exact point-to-point continuity with the earlier Python pipeline, but in return it produces a simulation workflow that is compatible with SUMO's modelling assumptions, and faithful to the calibrated locality-level demand structure.

— TODO could also compare to what infrastructure changes are already planned.

TODO Sustainable Urban Mobility Plan (SUMP) malta

TODO discuss data limits such as lack of on street parking data, or street width

5 Evaluation

6 Conclusion

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